

Build a Knowledge Graph

LESSON 02 · GROUND IN KNOWLEDGE – STUDIO: KNOWLEDGE GRAPH & SEMANTIC MODEL

What you'll learn

- Set up a Neo4j Aura Graph Database instance and extract its connection credentials (URI, username, password).
- Connect a Neo4j Graph Database to Lyzr Studio under Data Connections.
- Create a new Knowledge Graph in Studio and ingest unstructured text documents to automatically extract entities and relationships.
- Test knowledge graph retrieval and visualize the compiled graph directly inside the Studio interface.
- Configure retrieval strategies like basic similarity, MMR (Maximal Marginal Relevance), or Hyde (Hypothetical Document Embeddings).

Key concepts

A Knowledge Graph in Studio requires a connected graph database like Neo4j. Under the hood, Lyzr writes structural graph data to Neo4j. Users can bring their own hosted databases, such as Neo4j Aura (which has a free tier), to store these graphs.

Lyzr automates entity and relationship extraction during document ingestion. When a document is added to a Knowledge Graph, Lyzr reads the text, extracts key entities (things) and relationships (how they connect), and constructs the Neo4j graph nodes and edges automatically.

The graph database is an overlay, not a replacement for traditional RAG. A Knowledge Graph still supports passage retrieval for standard queries. The graph exists as an extra intelligence layer to answer connection-heavy questions.

Visualizing the graph helps verify the relationships extracted from your documents. The "Visualize Knowledge Graph" option in Studio renders your knowledge as a cluster map, letting you see dots (entities) and lines (relationships) that were parsed from the raw text.

Retrieval behavior can be customized using different retrieval algorithms. You can stick to defaults or toggle between similarity search, MMR (for diverse result sets), and Hyde (for generating hypothetical query-answers first).

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In Studio

Studio feature: **Connections** → **Data Connections** and **Build** → **Knowledge** → **Knowledge Graph**. Here's how to build, connect, and visualize a knowledge graph:

1. Go to **Connections**, click **Data Connections**, and choose **Add Connection**. Select **Neo4j**, paste your connection URI, username, and password, and save.
2. Go to **Build** and open the **Knowledge** screen. Click the **Knowledge Graph** tab.
3. Click **New Graph**, name it, choose your connected Neo4j database, and click **Create**.
4. Inside your new graph, click **Add Document** and upload a text file. Wait for the graph builder to read, parse, and write nodes to Neo4j (takes 1–2 minutes).
5. Open the retrieval tester drawer on the right to run test queries. Adjust search behaviors (Similarity, MMR, Hyde) under **Configure** if needed.
6. Click **Visualize Knowledge Graph** to open the interactive node-and-link map.

TRY THIS

- 1** Create a free Neo4j AuraDB instance, save the credentials, and hook it up to Lyzr Studio.
- 2** Upload a simple policy text document into your new Knowledge Graph in Studio. Open **Visualize Knowledge Graph** and trace the connection between two concepts (e.g., a warranty and its exclusion rules).
- 3** Try toggling between Similarity, MMR, and Hyde in the configuration drawer. Run the same test query and compare the retrieved passages.